Module 4

Lab - Reading Files (External resource)

File: example.txt:

This is line 1

This is line 2

This is line 3

Code:

```
from pyodide.http import pyfetch

import pandas as pd

filename="https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBMDeveloperSkillsNetwork-PY0101EN-SkillsNetwork/labs/Module%204/data/example1.txt"

async def download(url, filename):

    response = await pyfetch(url)

    if response.status == 200:

        with open(filename, "wb") as f:

            f.write(await response.bytes())

await download(filename, "example1.txt")

print("done")
```

Output:

Done

Code:

```
example1 = "example1.txt"

file1 = open(example1, "r")
```

Output:

There is no output for this code.

Code:

```
file1.name
```

Output:

'example1.txt'

Code:

```
file1.mode
```

Output:

'r'

Code:

```
FileContent = file1.read()
FileContent
```

Output:

'This is line 1 \nThis is line 2\nThis is line 3'

Code:

```
print(FileContent)
```

Output:

This is line 1

This is line 2

This is line 3

Code:

```
type(FileContent)
```

Output:

str

Code:

```
file1.close()
```

Output:

No output

Code:

```
with open(example1, "r") as file1:
    FileContent = file1.read()
```

```
    print(FileContent)
```

Output:

This is line 1

This is line 2

This is line 3

Code:

```
file1.closed
```

Output:

True

Code:

```
print(FileContent)
```

Output:

This is line 1

This is line 2

This is line 3

Code:

```
with open(example1, "r") as file1:
    print(file1.read(4))
```

Output:

This

Code:

```
with open(example1, "r") as file1:
    print(file1.read(4))
    print(file1.read(4))
    print(file1.read(7))
    print(file1.read(15))
```

Output:

This

is

line 1

This is line 2

Code:

```
with open(example1, "r") as file1:
    print(file1.read(16))
    print(file1.read(5))
    print(file1.read(9))
```

Output:

This is line 1

This

is line 2

Code:

```
with open(example1, "r") as file1:
    print("first line: " + file1.readline())
```

Output:

first line: This is line 1

Code:

```
with open(example1, "r") as file1:
    print(file1.readline(20))
    print(file1.read(20))
```

Output:

This is line 1

This is line 2

This

Code:

```
with open(example1,"r") as file1:
        i = 0;
```

```
    for line in file1:
        print("Iteration", str(i), ": ", line)
        i = i + 1
```

Output:

Iteration 0 : This is line 1

Iteration 1 : This is line 2

Iteration 2 : This is line 3

Code:

```
with open(example1, "r") as file1:
    FileasList = file1.readlines()
```

Output:

No output

Code:

```
FileasList[0]
```

Output:

'This is line 1 \n'

Code:

```
FileasList[1]
```

Output:

'This is line 2\n'

Code:

```
FileasList[2]
```

Output:

'This is line 3'

Lab - Writing Files

Code:

```
exmp2 = '/Example2.txt'
with open(exmp2, 'w') as writefile:
```

```
    writefile.write("This is line A")
```

Output:

No output

Code:

```
with open(exmp2, 'r') as testwritefile:
    print(testwritefile.read())
```

Output:

This is line A

Code:

```
with open(exmp2, 'w') as writefile:
    writefile.write("This is line A\n")
    writefile.write("This is line B\n")
```

Output:

No output

Code:

```
with open(exmp2, 'r') as testwritefile:
    print(testwritefile.read())
```

Output:

This is line A

This is line B

Code:

```
Lines = ["This is line A\n", "This is line B\n", "This is line C\n"]
Lines
```

Output:

['This is line A\n', 'This is line B\n', 'This is line C\n']

Code:

```
with open('/Example2.txt', 'w') as writefile:
    for line in Lines:
```

```
        print(line)
        writefile.write(line)
```

Output:

This is line A

This is line B

This is line C

Code:

```
with open('/Example2.txt', 'r') as testwritefile:
    print(testwritefile.read())
```

Output:

This is line A

This is line B

This is line C

Code:

```
with open('/Example2.txt', 'w') as writefile:
    writefile.write("Overwrite\n")
with open('/Example2.txt', 'r') as testwritefile:
    print(testwritefile.read())
```

Output:

Overwrite

Code:

```
with open('/Example2.txt', 'a') as testwritefile:
    testwritefile.write("This is line C\n")
    testwritefile.write("This is line D\n")
    testwritefile.write("This is line E\n")
```

Output:

No output

Code:

```
with open('/Example2.txt', 'r') as testwritefile:
    print(testwritefile.read())
```

Output:

```
Overwrite
This is line C
This is line D
This is line E
```

Code:

```
with open('/Example2.txt', 'a+') as testwritefile:
    testwritefile.write("This is line E\n")
    print(testwritefile.read())
```

Output:

No output

Code:

```
with open('/Example2.txt', 'a+') as testwritefile:
    print("Initial Location: {}".format(testwritefile.tell()))
    data = testwritefile.read()
    if (not data):                    #empty strings return false in python
        print('Read nothing')
    else:
        print(testwritefile.read())
    testwritefile.seek(0,0)           # move 0 bytes from beginning.
    print("\nNew Location : {}".format(testwritefile.tell()))
    data = testwritefile.read()
    if (not data):
        print('Read nothing')
    else:
```

```
        print(data)
    print("Location after read: {}".format(testwritefile.tell()) )
```

Output:

Initial Location: 85

Read nothing

New Location : 0

Overwrite

This is line C

This is line D

This is line E

This is line E

This is line E

Location after read: 85

Code:

```
with open('/Example2.txt', 'r+') as testwritefile:
    testwritefile.seek(0,0) #write at beginning of file
    testwritefile.write("Line 1" + "\n")
    testwritefile.write("Line 2" + "\n")
    testwritefile.write("Line 3" + "\n")
    testwritefile.write("Line 4" + "\n")
    testwritefile.write("finished\n")
    testwritefile.seek(0,0)
    print(testwritefile.read())
```

Output:

Line 1

Line 2

Line 3

Line 4

finished

D

This is line E

This is line E

This is line E

Code:

```
with open('/Example2.txt', 'r+') as testwritefile:
    testwritefile.seek(0,0) #write at beginning of file
    testwritefile.write("Line 1" + "\n")
    testwritefile.write("Line 2" + "\n")
    testwritefile.write("Line 3" + "\n")
    testwritefile.write("Line 4" + "\n")
    testwritefile.write("finished\n")
    #Uncomment the line below
    testwritefile.truncate()
    testwritefile.seek(0,0)
    print(testwritefile.read())
```

Output:

Line 1

Line 2

Line 3

Line 4

finished

Code:

```
with open('/Example2.txt','r') as readfile:
    with open('/Example3.txt','w') as writefile:
          for line in readfile:
                writefile.write(line)
```

Output:

No output

Code:

```
with open('/Example3.txt','r') as testwritefile:
    print(testwritefile.read())
```

Output:

Line 1

Line 2

Line 3

Line 4

finished

Exercise

Code:

```
from random import randint as rnd
memReg = '/members.txt'
exReg = '/inactive.txt'
fee =('yes','no')
def genFiles(current,old):
    with open(current,'w+') as writefile:
        writefile.write('Membership No  Date Joined  Active  \n')
        data = "{:^13}  {:<11}  {:<6}\n"

        for rowno in range(20):
            date = str(rnd(2015,2020))+ '-' + str(rnd(1,12))+'-'+str(rnd(1,25))
            writefile.write(data.format(rnd(10000,99999),date,fee[rnd(0,1)]))
 with open(old,'w+') as writefile:
        writefile.write('Membership No  Date Joined  Active  \n')
        data = "{:^13}  {:<11}  {:<6}\n"
```

```
        for rowno in range(3):
            date = str(rnd(2015,2020))+ '-' + str(rnd(1,12))+'-'+str(rnd(1,25))
            writefile.write(data.format(rnd(10000,99999),date,fee[1]))
genFiles(memReg,exReg)
```

Output:

No output

Output:

```
def testMsg(passed):
    if passed:
       return 'Test Passed'
    else :
       return 'Test Failed'

testWrite = "/testWrite.txt"
testAppend = "/testAppend.txt"
passed = True

genFiles(testWrite,testAppend)

with open(testWrite,'r') as file:
    ogWrite = file.readlines()

with open(testAppend,'r') as file:
    ogAppend = file.readlines()

try:
    cleanFiles(testWrite,testAppend)
except:
```

```
    print('Error')


with open(testWrite,'r') as file:
    clWrite = file.readlines()


with open(testAppend,'r') as file:
    clAppend = file.readlines()


# checking if total no of rows is same, including headers


if (len(ogWrite) + len(ogAppend) != len(clWrite) + len(clAppend)):
    print("The number of rows do not add up. Make sure your final files have the same header and format.")
    passed = False


for line in clWrite:
    if  'no' in line:
        passed = False
        print("Inactive members in file")
        break
    else:
        if line not in ogWrite:
            print("Data in file does not match original file")
            passed = False
print ("{}".format(testMsg(passed)))
```

Output:

Inactive members in file

Test Failed

Code:

```
%pip install xlrd openpyxl
```

Output:

No output

Code:

```
import pandas as pd
```

Output:

No output

Code:

```
from pyodide.http import pyfetch

import pandas as pd

filename = "https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/LXjSAttmoxJfEG6il1Bqfw/Product-sales.csv"


async def download(url, filename):

    response = await pyfetch(url)

    if response.status == 200:

        with open(filename, "wb") as f:

            f.write(await response.bytes())

await download(filename, "Product-sales.csv")

df = pd.read_csv("Product-sales.csv")
```

Output:

No output

Code:

```
df.head()
```

Output:

	OrderID	Product	Category	Quantity	Price	Total	OrderDate	CustomerCity
0	1	Laptop	Electronics	2	800	1600	2022-01-10	New York
1	2	Smartphone	Electronics	3	600	1800	2022-02-15	Los Angeles
2	3	Desk Chair	Furniture	5	150	750	2022-03-12	Chicago
3	4	Notebook	Stationery	10	2	20	2022-04-05	Houston
4	5	Monitor	Electronics	1	300	300	2022-05-21	Miami

Code:

xlsx_path = 'https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/n9LOuKI9SlUa1b5zkaCMeg/Product-sales.xlsx'

await download(xlsx_path, "Product-sales.xlsx")

df = pd.read_excel("Product-sales.xlsx")

df.head()

Output:

	OrderID	Product	Category	Quantity	Price	Total	OrderDate	CustomerCity
0	1	Laptop	Electronics	2	800	1600	2022-01-10	New York
1	2	Smartphone	Electronics	3	600	1800	2022-02-15	Los Angeles
2	3	Desk Chair	Furniture	5	150	750	2022-03-12	Chicago
3	4	Notebook	Stationery	10	2	20	2022-04-05	Houston
4	5	Monitor	Electronics	1	300	300	2022-05-21	Miami

Code:

x = df[['Quantity']]

x

Output:

	Quantity
0	2
1	3
2	5
3	10
4	1

Code:

x = df['Product']

x

Output:

0 Laptop

1 Smartphone

2 Desk Chair

3 Notebook

4 Monitor

Name: Product, dtype: object

Code:

x = df[['Quantity']]

type(x)

Output:

pandas.core.frame.DataFrame

Code:

y = df[['Product','Category', 'Quantity']]

y

Output:

	Product	Category	Quantity
0	Laptop	Electronics	2
1	Smartphone	Electronics	3
2	Desk Chair	Furniture	5
3	Notebook	Stationery	10
4	Monitor	Electronics	1

Code:

df.iloc[0, 0]

Output:

1

Code:

```
df.iloc[1,0]
```

Output:

2

Code:

```
df.iloc[0,2]
```

Output:

'Electronics'

Code:

```
df.iloc[1,2]
```

Output:

'Electronics'

Code:

```
df.loc[0, 'Product']
```

Output:

'Laptop'

Code:

```
df.loc[1, 'Product']
```

Output:

'Smartphone'

Code:

```
df.loc[1, 'CustomerCity']
```

Output:

'Los Angeles'

Code:

```
df.loc[1, 'Total']
```

Output:

1800

Code:

df.iloc[0:2, 0:3]

Output:

	OrderID	Product	Category
0	1	Laptop	Electronics
1	2	Smartphone	Electronics

Code:

df.loc[0:2, 'OrderID':'Category']

Output:

	OrderID	Product	Category
0	1	Laptop	Electronics
1	2	Smartphone	Electronics
2	3	Desk Chair	Furniture

Quiz on DataFrame

Code:

q = df[['Price']]

q

Output:

	Price
0	800
1	600
2	150
3	2
4	300

Code:

```
q = df[['Product', 'Category']]
q
```

Output:

	Product	Category
0	Laptop	Electronics
1	Smartphone	Electronics
2	Desk Chair	Furniture
3	Notebook	Stationery
4	Monitor	Electronics

Code:

```
df.iloc[1, 2]
```

Output:

'Electronics'

Code:

```
df_new=df
df_new.index=new_index
df_new.loc['a', 'CustomerCity']
df_new.loc['a':'d', 'CustomerCity']
```

Output:

```
a       New York
b    Los Angeles
c        Chicago
d        Houston
Name: CustomerCity, dtype: object
```